

Low-resource biometric speaker verification on Sinhala–Tamil bilingual audio: a single-corpus ablation study with cross-lingual transfer, score normalisation, and PLDA scoring

[First Author]¹ and [Second Author]¹

¹[Department], [University], Sri Lanka. , [email@institution]

Abstract

Voice-based biometric authentication has matured on English benchmarks but remains under-evaluated for the low-resource languages typical of South Asian deployment contexts. We present the first reproducible speaker-verification (SV) benchmark for Sri Lankan Sinhala and Tamil, evaluated on a two-tier protocol: benchmark v0, 527 speakers of read speech from OpenSLR SLR52 and SLR65 (478 Sinhala, 49 Tamil; 12,000 trials per language), and benchmark v1, built on SLCeleb, our 280-speaker in-the-wild corpus (pending data placement). Building on an ECAPA-TDNN encoder with AAM-Softmax training, we systematically evaluate four authentication-relevant levers: cross-lingual fine-tuning from an English-pretrained checkpoint with selective front-end freezing; layer-wise learning-rate decay; an auxiliary language-identification head trained jointly with the speaker loss; and a two-stage PLDA followed by adaptive symmetric score normalisation. Each lever is ablated independently, with three- seed reproducibility. We report monolingual Sinhala-only, Tamil-only, and cross-lingual equal-error rates, together with an open-source toolkit and corpus-preparation pipeline released for downstream use.

Keywords: speaker verification; biometrics; cross-lingual transfer; low-resource speech; Sinhala; Tamil; PLDA; score normalisation; ECAPA-TDNN.

1 Introduction

Voice-based biometrics is increasingly deployed in financial authentication, call-centre fraud detection, and forensic audio analysis. The dominant published benchmarks [5] use English speech almost exclusively, and the state-of-the-art architectures and training recipes [2, 1] were validated at that scale. For Sri Lankan deployment contexts the linguistic mismatch is fundamental: Sinhala (Indo-Aryan, [N=17] million speakers) and Tamil (Dravidian, [N=5] million Sri Lankan speakers) are absent from the publicly available SV corpora, bilingual and code-switched speech is normal in everyday usage, and the labelled-speaker base accessible to a single research effort is small (tens to low hundreds).

This paper contributes:

1. A reproducible two-tier Sinhala/Tamil SV benchmark — v0: 527 speakers (478 Sinhala from OpenSLR SLR52, 49 Tamil from OpenSLR SLR65 read speech); v1: SLCeleb, our 280-speaker in-the-wild corpus (pending) — with monolingual, cross-lingual, and pooled trial lists.
2. Single-feature ablations isolating the contribution of cross-lingual fine-tuning, layer-wise LR decay, multi-task language-ID, AS-Norm score normalisation, and PLDA scoring.

3. An open-source toolkit (`voxceleb_trainer-SL-SPV`) [6] extending the upstream `voxceleb_trainer` with eleven features and twenty-six bug fixes, each selectable via a single YAML key.

The remainder of this paper is organised as follows. Section 2 reviews relevant work in SV architectures, cross-lingual transfer, and score normalisation. Section 3 describes the corpus, architectures, and training recipe. Section 4 reports results. Section 5 discusses biometric-deployment implications, and Section 6 concludes.

2 Related work

2.1 Speaker verification architectures

Modern SV pipelines extract a fixed-length embedding from variable-length audio and score trial pairs by a similarity function on those embeddings. ECAPA-TDNN [2] is the de-facto reference architecture, combining Res2Net-style multi-scale 1D convolutions, squeeze-and-excite channel re-weighting, multi-layer feature aggregation, and channel-attentive statistical pooling. We adopt ECAPA-TDNN-Large ($C = 1024$ channels) as the headline encoder.

2.2 Cross-lingual transfer for biometrics

When the target-language SV corpus is small, training from random initialisation overfits. Cross-lingual fine-tuning from a high-resource (typically English VoxCeleb1) checkpoint is the established remedy. The fine-tune policy hyper-parameters — which layers to freeze, the global LR multiplier, and whether to apply layer-wise LR decay (LLRD) — are well studied in the BERT / WavLM literature but less systematically reported for SV fine-tuning, which motivates our explicit ablation in Section 4.

2.3 Score normalisation and generative backends

Adaptive symmetric score normalisation (AS-Norm) [4] computes per-trial-side cohort statistics from a fixed set of impostor embeddings and normalises trial scores against those statistics. Two-covariance PLDA [7] replaces cosine scoring with a likelihood ratio test under same-speaker vs different-speakers hypotheses. Both are eval-time post-processing steps that do not require retraining, and they compose: the standard NIST SRE pipeline order is PLDA $\rightarrow$ AS-Norm.

2.4 Auxiliary objectives for low-resource SV

Multi-task language-ID heads — a linear classifier trained jointly with the speaker loss to predict the utterance’s language — regularise the speaker embedding to retain language-discriminative structure that aids cross-lingual discrimination. The opposite (adversarial / DANN-style [3]) is also studied but empirically more brittle on small corpora; we therefore evaluate the multi-task variant as the primary lever and treat DANN as infrastructure-ready future work.

3 Methodology

3.1 Corpus

Evaluation is two-tier. Benchmark v0 comprises 527 speakers: 478 Sinhala from OpenSLR SLR52 (crowdsourced read speech, 16 kHz) and 49 Tamil from OpenSLR SLR65 (studio read speech, resampled to 16 kHz at load), keeping speakers with at least 10 utterances. Benchmark v1 is built on SLCeleb, our 280-speaker in-the-wild Sinhala/Tamil corpus (multi-genre, multi-session, with bilingual speakers), pending data placement. Within each speaker we hold out 20% of

utterances for trial-pair construction; the remainder forms the training pool (31,049 training utterances at a 60-utterance-per-speaker cap for the fine-tuning experiments).

v0 trial pairs are constructed at a 1:3 target-to-impostor ratio at three operating points: monolingual Sinhala (3,000 target + 9,000 impostor pairs), monolingual Tamil (3,000 + 9,000), and cross-lingual (pairs for speakers present in both languages — empty at v0, since the OpenSLR corpora contain no bilingual speakers, and populated at v1 by SLCeleb). Pair sampling is reproducible from a single fixed random seed [6].

3.2 Architecture and training

The speaker encoder is ECAPA-TDNN [2] with $C = 1024$ channels, 80 log-mel filterbanks, embedding dimension 192, and channel-attentive statistical pooling. The training loss is additive-angular-margin softmax (AAM-Softmax) [1] with margin 0.2 and scale 30. Augmentation reuses the MUSAN + RIR simulation chain at training time.

3.3 Cross-lingual fine-tune protocol

For configurations P1 and P2 (Section 4) we initialise from an ECAPA-TDNN checkpoint pretrained on VoxCeleb1 [5]. During SL fine-tuning the mel front-end is frozen, the global LR is scaled by $0.1\times$, and layer-wise LR decay with $\gamma = 0.9$ is applied across the four ECAPA layers so that the layer closest to the loss head receives the full LR and the input-side layer receives $0.9^3 \approx 0.73\times$ of it.

3.4 Multi-task language-ID head

A linear head $W_{\text{aux}} \in \mathbb{R}^{4 \times 192}$ predicts the utterance’s primary language (si / ta / en / mix) from the speaker embedding before the $n_{\text{PerSpeaker}}$ reshape. The combined loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{speaker}} + \lambda \mathcal{L}_{\text{lang}}, \quad \lambda = 0.3. \quad (1)$$

Speakers absent from the per-speaker language lookup carry `ignore_index=-1` and contribute no gradient.

3.5 Eval-time scoring backends

We evaluate three eval-time scoring backends:

1. Cosine similarity on length-normalised embeddings (baseline).
2. Two-covariance PLDA [7] with LDA pre-reduction to $d = 200$ and length normalisation.
3. AS-Norm [4] with top- $K = 300$ cohort statistics on a 500-speaker in-domain cohort, layered on top of either backend (1) or (2).

3.6 Reproducibility

All experiments use deterministic-mode training (cuDNN deterministic flag, TF32 disabled, `torch.use_deterministic_algorithms` enabled). Primary configurations are run with three random seeds (42, 123, 7); we report mean $\pm$ std. Single-feature ablations use seed 42 only.

4 Experiments

4.1 Configurations

We define three primary configurations and five single-feature ablations:

Table 1: Preliminary monolingual EER (%) on benchmark v0 (527 read-speech speakers; 12,000 trials per language). Fine-tuned rows are mean $\pm$ std over three seeds. Preliminary, benchmark v0: read speech, near single-session, closed-set speakers for the fine-tuned rows; v1 (SLCeleb) provides the open-set, in-the-wild condition.

System	Si	Ta
Zero-shot SpeechBrain ECAPA-TDNN	4.78	3.21
Zero-shot ReDimNet-B6	2.71	1.48
ReDimNet-B6 + AS-Norm (training-free)	2.07	0.90
WavLM-base+ LoRA fine-tune (3 seeds)	3.73 ± 0.49	3.11 ± 0.61
WavLM-base+ full fine-tune (3 seeds)	1.69 ± 0.06	1.41 ± 0.10

Table 2: Speaker-verification performance on the SL-SPV evaluation set. Numbers are pooled / Sinhala-only / Tamil-only / cross-lingual EER (%), mean $\pm$ std over three random seeds.

Config	Pooled	Si	Ta	Cs
P0 from scratch	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$
P1 + fine-tune	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$
P2 full stack	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$	$[X.X \pm Y.Y]$

- **P0:** ECAPA-TDNN from scratch on SL-SPV.
- **P1:** P0 + cross-lingual fine-tune (front-end frozen, LLRD).
- **P2:** P1 + multi-task language-ID head + AS-Norm + PLDA + per-language evaluation.
- Ablations of P2 with each of the five features individually disabled.

4.2 Preliminary monolingual baselines (benchmark v0)

Table 1 gives preliminary monolingual context from benchmark v0. Deployment-relevant preliminary findings: full fine-tuning beat LoRA at every seed on this closed-set benchmark (contrary to the published PEFT-under-shift prediction; the v1 open-set rerun arbitrates), and per-language score calibration is mandatory — applying a calibrator fitted on one language to the other inflates Cllr by up to $5.7\times$. Cross-lingual (Cs) trials remain pending on benchmark v1’s bilingual speakers.

4.3 Headline results

4.4 Ablation analysis

5 Discussion

5.1 Implications for biometric deployment

Voice biometric deployment in bilingual environments faces three operating points that this benchmark explicitly characterises: monolingual enrolment matching the test-time language (Si / Ta columns of Table 2), cross-lingual enrolment-test mismatch (Cs column), and pooled (deployment-time language unknown). A smaller Si–Ta gap indicates a more deployable model when the language is not known in advance; a smaller Cs–Pooled gap indicates robustness to language switching between enrolment and verification.

Table 3: Single-feature ablations from the full stack (P2), seed 42. Δ = pooled-EER increase when the named feature is removed.

Ablation	Pooled EER %	Δ
P2 (all features)	[X.X]	0
– fine-tune	[X.X]	+ [Y.Y]
– LLRD	[X.X]	+ [Y.Y]
– lang-aux	[X.X]	+ [Y.Y]
– AS-Norm	[X.X]	+ [Y.Y]
– PLDA	[X.X]	+ [Y.Y]

5.2 Threats to validity

[NOTE: Discuss after results land. Likely topics: benchmark v0’s read-speech, near single-session character and closed-set fine-tuning evaluation (optimistic absolute EERs; v1 SLCeleb provides the open-set, multi-session condition), single-corpus-per-language generalisation, channel homogeneity, and possible label noise in the multi-task language-ID lookup.]

5.3 Future work

The toolkit ships infrastructure for two avenues we did not exercise in this paper: (i) self-supervised continued pretraining on unlabelled SL podcast/broadcast audio (FEATURE-009, design-only) to further close the cross-lingual gap; and (ii) DANN-style adversarial language and channel invariance (FEATURE-008), particularly relevant for telephony deployment where the recording channel differs from the training distribution.

6 Conclusion

We presented the first reproducible speaker-verification benchmark for Sri Lankan Sinhala and Tamil, evaluated on a two-tier protocol (benchmark v0: 527 read-speech speakers from OpenSLR SLR52/SLR65; benchmark v1: SLCeleb, our 280-speaker in-the-wild corpus, pending). Single-feature ablations isolate the contribution of each lever in the full-stack configuration: cross-lingual fine-tune, layer-wise LR decay, multi-task language-ID, AS-Norm, and PLDA. The open-source toolkit, corpus-preparation pipeline, and reproducibility checklist are released for downstream work on Sri Lankan biometrics.

Acknowledgements

[Funding, computational resources, advisor acknowledgements.]

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Biographical notes

[First Author] is [position] at [institution]. Their research interests include [topics]. [One-paragraph bio per author per IJBM submission guidelines.]